1.

(a) $\frac{1}{3}\left(\begin{bmatrix}0\\10\end{bmatrix}+\begin{bmatrix}0\\5\end{bmatrix}+\begin{bmatrix}0\\0\end{bmatrix}\right)=\begin{bmatrix}0\\5\end{bmatrix}$

(b) $\Sigma = cov(X) = \frac{1}{n-1} X X^{T}$

$$= \frac{1}{2}\begin{bmatrix}0 & 0 & 0\\ 5 & 0 & -5\end{bmatrix}\begin{bmatrix}0 & 5\\ 0 & 0\\ 0 & -5\end{bmatrix}$$

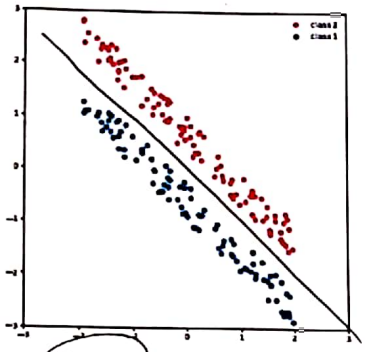
$$= \begin{bmatrix}0 & 0\\ 0 & 25\end{bmatrix}$$

(c) ~~the left, upper index in the matrix signifies~~ the left, upper index (1,1) in the matrix signifies the x variance, the right, bottom (2,2) index in the matrix signifies y variance. No change in data in x direction, so, it's 0 variance. but there are 2 variances from the mean, 5, in y direction, $\sigma_1=(10-5)^2$, $\sigma_2=(5-0)^2$, as 25.

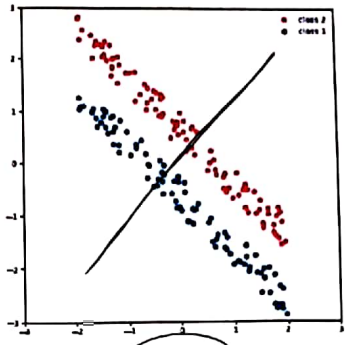
$\therefore \sigma_{22}=25$, $\sigma_{11}=0$.

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#2

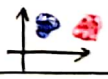


PCA



LDA

(a) ⇒ don't have labels to supervise
⇒ don't know shape ~~or~~ colors (shape or color)
⇒ treat each data equally
⇒ want to find direction with great variance

perpendicular to PCA direction
⇒ need to take care of different classes
⇒ projecting ~~data~~ in the direction will classify them into 2 classes based on classes
⇒ best way to classify/discriminate one another in the direction: 

(b) LDA!

As discussed in (a) LDA part, and since it is binary classification (0 or 1 / class 1 or class 2), LDA is best method for binary classification on ~~data~~ in the reduced 1-D space. It is because LDA projects/classifies the data into 2 classes much better than PCA

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#3 (a), and (c)

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